1. $I = \iint\limits_D \sqrt{1-x^2-y^2}\,\mathrm{d}\sigma$, where D is the region bounded by the curve $y = \sqrt{1-x^2}$ and the line $y = 0$.

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2. Sketch each region of integration and change the order of integration in each iterated integral: (1) $\int_1^2 dx \int_{2-x}^{\sqrt{2x-x^2}} f(x,y)dy$ (2) $\int_1^2 dx \int_{\frac{1}{x}}^1 f(x,y)dy$
3. Evaluate the following double integrals: (1) $\iint\limits_D (3x-2y)\,d\sigma$, where D is the closed region bounded by the two coordinate axes and the line $x+y=2$ (2) $\iint\limits_D xy^2\,d\sigma$, where D is the right half of the closed disk $x^2+y^2=4$ bounded by the y axis
4. Find the volume of the solid bounded by the coordinate planes, the planes $x=4$ and $y=4$, and the paraboloid $z = x^2+y^2+1$.
5. Evaluate the following double integrals using polar coordinates: (1) $\iint\limits_D e^{-x^2-y^2}\,\mathrm{d}\sigma$, where D is the part of the unit disk in the first quadrant

(2) $\iint\limits_D (4-x-y)\,\mathrm{d}\sigma$, where D is the disk $x^2+y^2 \le 2y$

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6. Find the area of the part of the hemisphere $z = \sqrt{2-x^2-y^2}$ cut off by the cone $z = \sqrt{x^2+y^2}$ that lies inside the cone.
7. Find the surface area of the solid bounded by two mutually perpendicular circular cylinders $x^2+y^2 = R^2$ and $x^2+z^2 = R^2$ of the same base radius.